

Cancer and its treatments – an overview

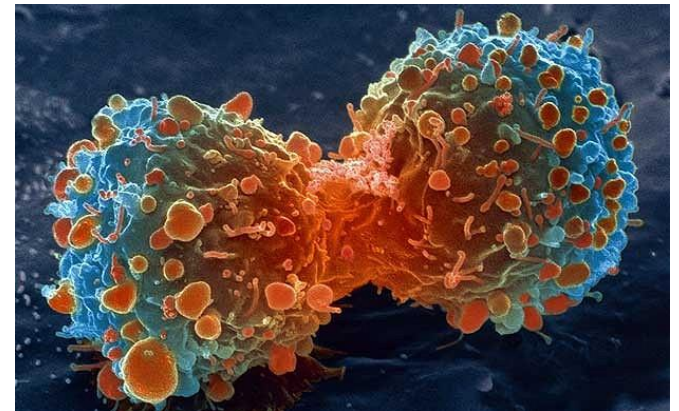
Lucy Jagger MA PgDip RN BA(Hons)

Insurance Nurse Specialist, Unum

May 2014

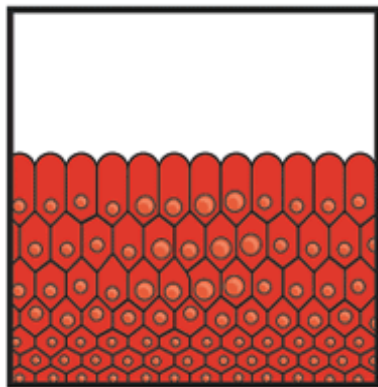
Aim of session

- What is cancer?
- What causes cancer?
- Statistics
- Overview of treatments
- Chemotherapy
- Hormone Therapies
- Biological Therapies
- Radiotherapy

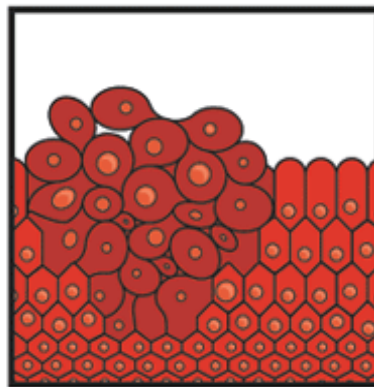


What is cancer?

Cancer is a class of diseases characterised by uncontrolled cell division and the ability to invade other tissues either by invasion or migration.



Normal cells



Cells forming a tumour

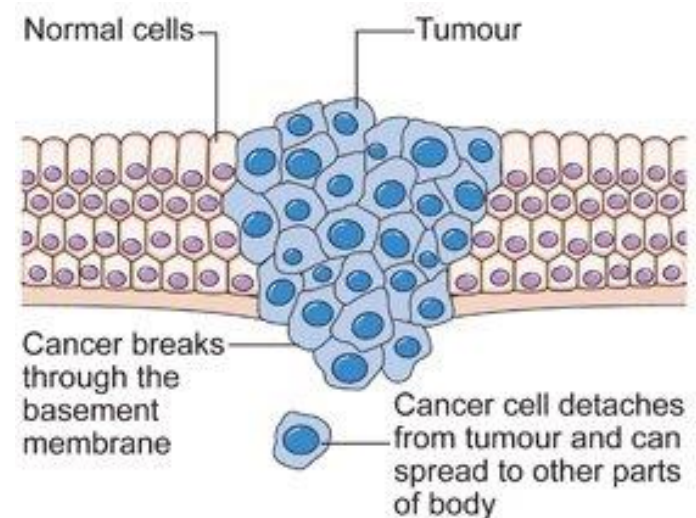


Diagram showing a malignant tumour
Copyright © CancerHelp UK

What is metastasis?

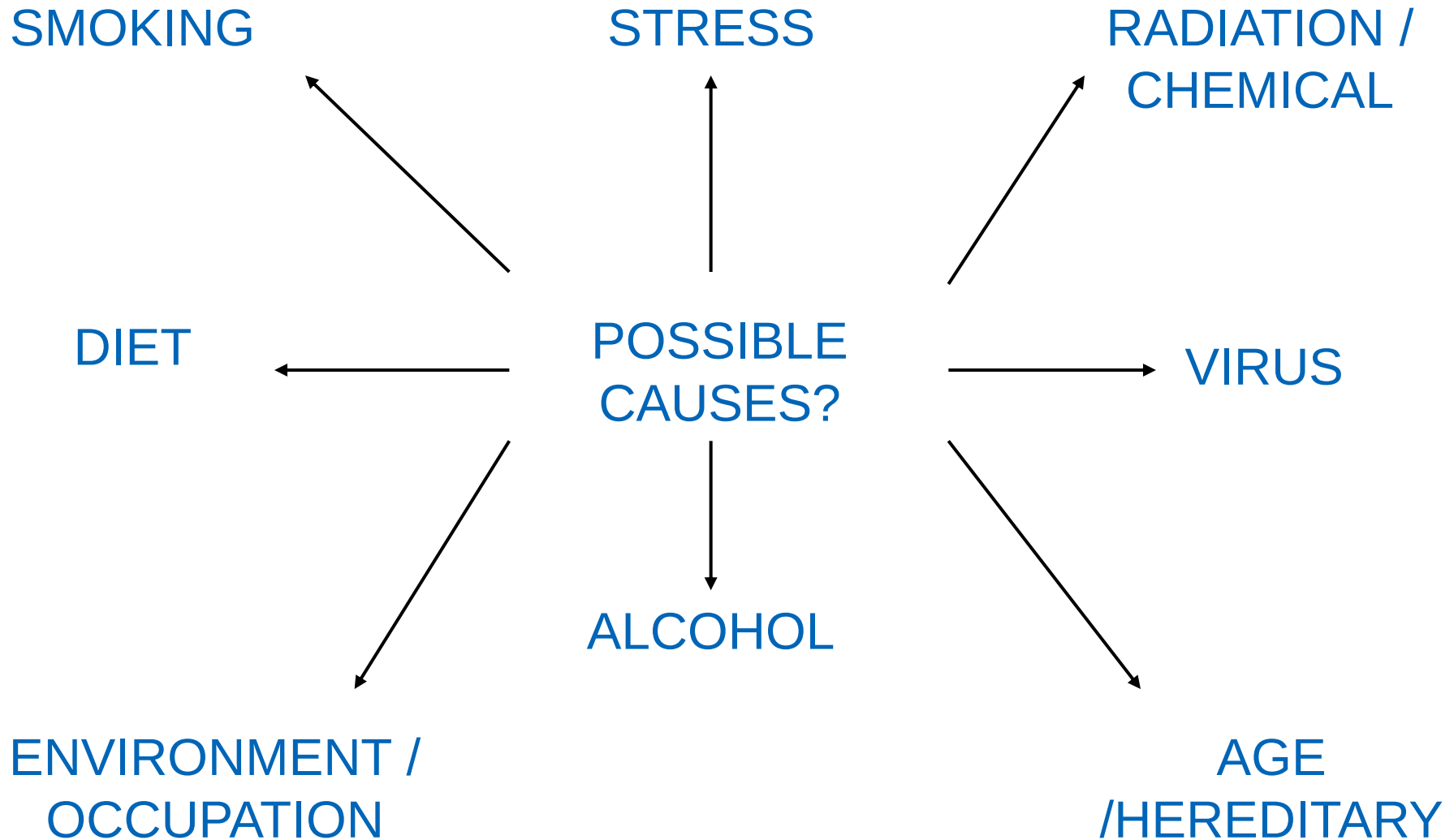
meta + *stasis*
change of *place*

- The spread of cancer from its primary site to other parts of the body
- All cancers can be metastatic
- Whether they are is dependent on the host environment

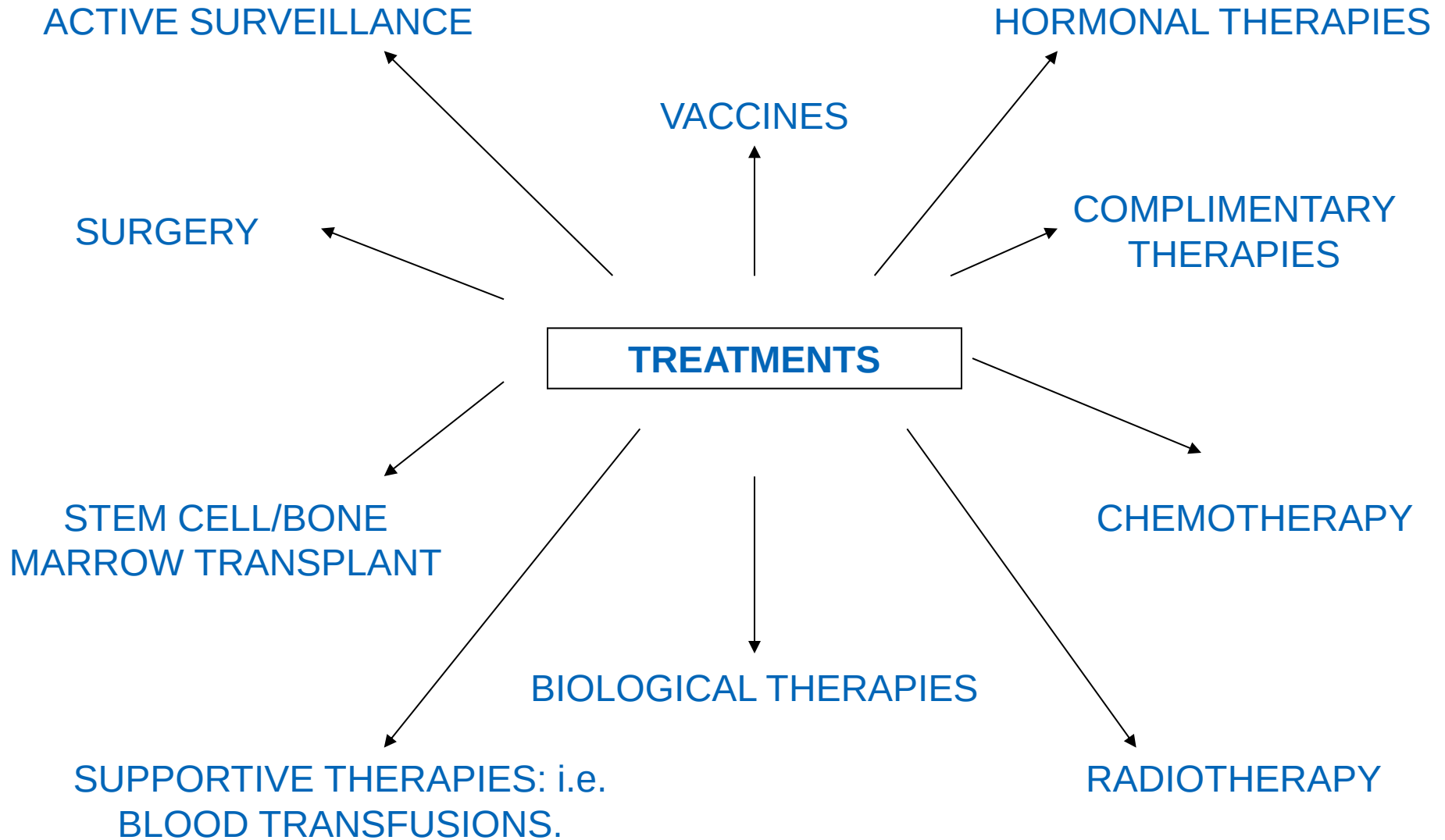
Statistics

- There are 200 different types of cancer
- Cancer has been with us throughout the history of mankind
- Second leading cause of death in the western world to heart disease
- 300,000 people each year are diagnosed with cancer in the UK (Macmillan)
- It has been estimated that more than 1 in 3 people (33%) will develop cancer at some point in their lifetime (Macmillan)
- Statistics in 2008 showed most common cancer for men was prostate cancer (24%) and for women was breast cancer (31%). The second most common for both men and women was lung cancer (Macmillan)
- On average a patient with cancer now has a 46.2% chance of being alive 10 years after diagnosis (Cancer research UK)
- The number of cancer survivors is increasing by 3.2% each year
- There are over three quarters of a million people of working age with a cancer diagnosis (Macmillan)

Possible causes



Treatments



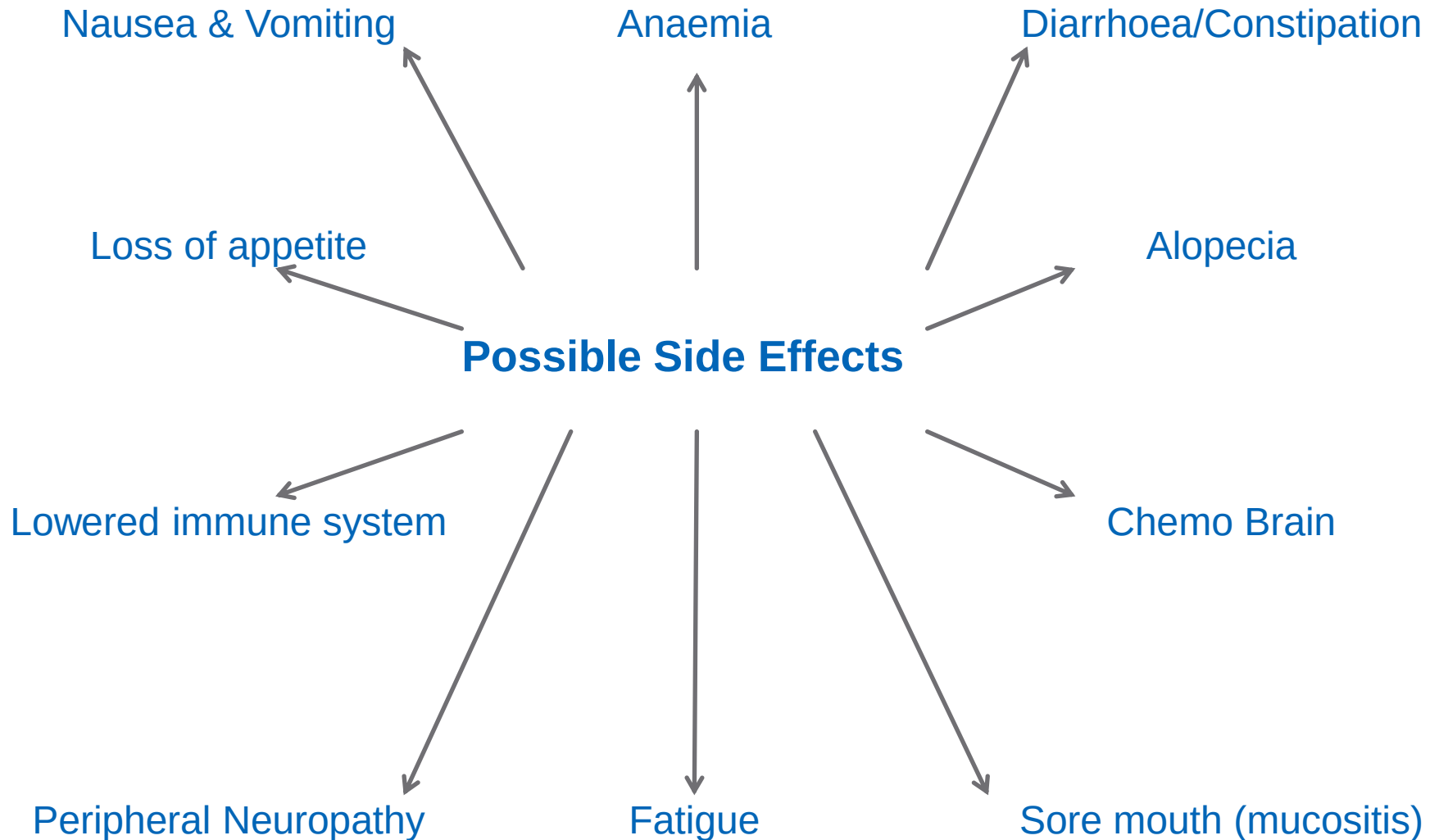
Chemotherapy

- Chemotherapy: treatment with powerful drugs that are most often given by mouth or by injections. Drugs are carried in the blood so can reach cancer cells anywhere in the body
- Aim of chemotherapy:** To achieve selective kill of cancer cells without killing normal cells
- Cytotoxic drugs interfere with protein synthesis and cell division:
 - DNA –determines the protein made by the cell and cell function (library of the cell)
 - RNA – receives instructions from DNA and makes proteins (reads the library)
- Cytotoxic drugs will target different areas of the cells cycle and damage cancer cells in different ways
- Chemotherapy is given in cycles each followed by a recovery period to allow time for the normal cells to recover.
- Chemotherapy combination regimen is often used as:**
 - Each cytotoxic drug is chosen for its different effects (mode of action)
 - Maximum cell kill and broad coverage against multiple cell lines
 - Different toxicities

Aims of chemotherapy

- **To cure cancer**
 - With some types of cancer, the aim is to destroy all the cancer cells and cure the disease, i.e. leukaemia
- **To reduce the risk of cancer coming back**
 - Chemotherapy or radiotherapy may be given after surgery in case any cancer cells remain
- **To shrink a cancer and prolong life**
 - If a cure is not possible, chemotherapy may be given to shrink and try to control a cancer and prolong a life (palliation).

Possible side-effects of chemotherapy



Chemotherapy – breast cancer

Aim: To reduce the risk of breast cancer coming back

Normally occurs following breast surgery but sometimes before surgery to shrink tumour (inflammatory breast cancer)

Normally minimum of 6 cycles of chemotherapy – duration 1-3 weeks

Common chemotherapy regimen:

•FEC (5-Fluorouracil, Epirubicin & Cyclophosphamide)

- 3 weekly cycle (Day 1 chemotherapy, Day 2-20 no chemotherapy)
- Specific side effects: Alopecia and Hand-Foot syndrome

•FEC-T (FEC followed by Docetaxel (Taxotere®))

- 3 weekly cycle (Day 1 chemotherapy, Day 2-20 no chemotherapy)
- Normally 3-4 cycles of FEC followed by 3-4 cycles of Docetaxel
- Specific side effects: Alopecia and Peripheral neuropathy

•EC (Epirubicin & Cyclophosphamide)

- 3 weekly cycle (Day 1 chemotherapy, Day 2-20 no chemotherapy)
- Specific side effects: Alopecia

Chemotherapy – bowel cancer

Aim: In people with stage 3 colon cancer, there is a greater risk that the cancer will come back after surgery, so chemotherapy is usually recommended

Common combination of the following chemotherapy drugs:

5-Fluorouracil (5FU), Capecitabine, Oxaliplatin & Irinotecan

Common combination chemotherapy regimen:

Oxaliplatin with 5FU & folonic acid (FOLFOX):

- 2 weekly cycles and up to 12 cycles (Day 1 – Injection of Oxaliplatin and folonic acid followed by an infusion of 5FU over 2 days)
- Specific side effects: Peripheral neuropathy and Hand-Foot syndrome

Oxaliplatin & Capecitabine:

- 3 weekly cycles – 4 – 8 cycles over 3 -9 months (Day 1 Oxaliplatin and Capecitabine, then take Capecitabine (oral tablet) until day 14 and then no chemotherapy for 7 days)
- Specific side effects: Peripheral neuropathy and Hand-Foot syndrome

•Irinotecan with 5FU & folonic acid (FOLFIRI):

- 2 weekly cycles and up to 12 cycles over 6 months (Day 1 – Injection of Irinotecan and folonic acid followed by an infusion of 5FU over 2 days)
- Specific side-effects: Diarrhoea and Hand-Foot syndrome

Hormonal therapies

- Hormones are substances that occur naturally in the body and act as chemical messengers, which influence the growth and activity of cells
- Cancers that are hormone sensitive or hormone dependent need hormones to grow or develop
- Hormonal therapies work by altering the production or activity of particular hormones in the body. They can slow down or stop the growth of cancer by either:
 - *Stopping hormones being made.*
 - *Preventing hormones from making cancer cells grow and divide.*
- Most commonly treat breast and prostate cancers

Hormonal therapies – breast cancer

- Hormonal therapies only work for woman who have oestrogen receptor positive cancers
- Hormonal therapies reduce the level of oestrogen in the body or prevent it from attaching to the cancer cells.
- Aim in Early Breast Cancer: To reduce the risk of the cancer coming back.
- Main Hormonal Therapies: Tamoxifen or Aromatase Inhibitors (Arimidex® or Letrozole®)

Hormonal therapies – breast cancer

Tamoxifen:

- It works by stopping oestrogen receptors from attaching to breast cancer cells and encouraging them to grow
- Tamoxifen blocks the receptors
- Both pre and postmenopausal women can take Tamoxifen
- Tamoxifen comes in a tablet form
- Most women take Tamoxifen for 5 years.
- Side-effects: Hot flushes (45%), Sweats, Joint pain, Tiredness (25%)
- Side-effects- more troublesome in the first few months and then gets better in time.



Hormonal therapies – breast cancer

Aromatase Inhibitors (Arimidex® & Letrozole®):

- Post menopausal woman
- After menopause, ovaries stop producing oestrogen but the body still makes a small amount by changing other hormones (androgens) into oestrogen through an enzyme called aromatase.
- Aromatase inhibitors block this enzyme so it can't change androgens into oestrogen.
- Aromatase Inhibitors comes in a tablet form.
- Most women take Aromatase Inhibitor for 5 years.
- Side-effects: Headaches, Hot flushes and Sweats, Joint pain (30%), osteoporosis, Mood changes (20%) & Fatigue (20%)
- Side-effects- more troublesome in the first few months and then gets better in time.



Biological therapies

- Drugs that mimic substances that occur naturally in the body to destroy cancer cells
- Biological therapies work by changing the way that cells interact or signal to each other.
- **Biological or targeted therapies can be used to:**
 - *stimulate the immune system to attack cancer cells*
 - *control the growth of cancer cells*
 - *to overcome side effects of treatment*
- **There are several types of biological treatment:**
 - Monoclonal antibodies
 - Cancer growth inhibitors
 - Angiogenesis inhibitors
 - Vaccines

Monoclonal antibodies (MAB's)

- Monoclonal antibodies also known as 'targeted therapies'
- Monoclonal antibodies are designed to recognise and attach to specific proteins on the surface of cells
- Each Monoclonal Antibody recognises one particular protein.
- They work in different ways depending on the protein they are targeting, therefore different monoclonal antibodies have to be made to target different types of cancer
- **Three main types of monoclonal antibodies:**
 - *Triggering the body's immune system to attack the cancer cell and cause the cell to kill itself*
 - *Block signals telling the cancer cells to divide*
 - *Carry cancer drugs or radiation to cancer cells*

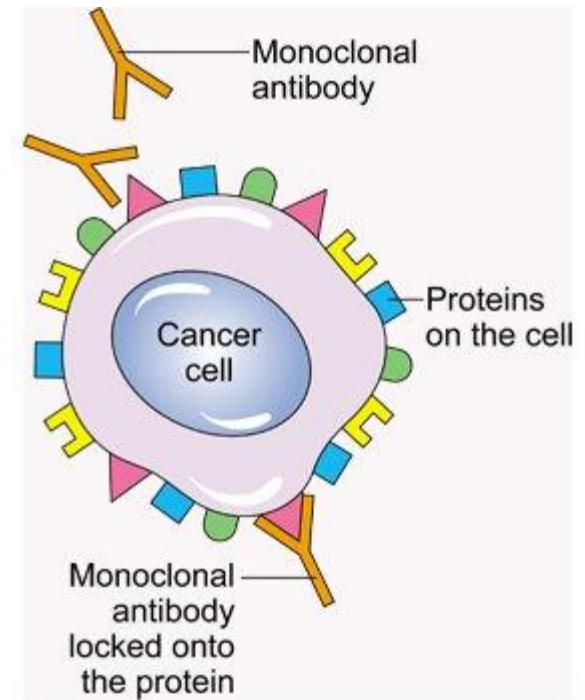
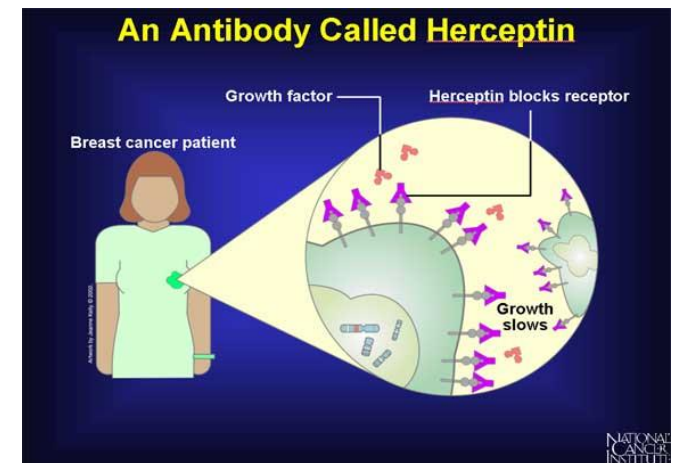


Diagram showing a monoclonal antibody attached to a cancer cell
© CancerHelp UK

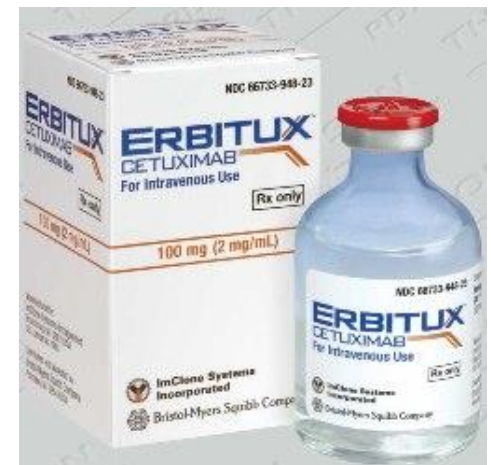
Trastuzumab (Herceptin®)

- Some cancers have too much a protein called human epidermal growth factor receptor 2 (HER 2) on the surface of their cells. They are called HER2 positive cancers
- The extra HER2 receptors stimulate the cancer cells to divide and grow
- Herceptin locks on to the HER2 proteins. This blocks the receptor and stops the cells from dividing and growing
- Herceptin only works in people who have cancer with high levels of the HER2 protein
- Licensed for HER2 positive early stage breast cancer and advanced stage breast and stomach (if the adenocarcinoma is HER2 positive).
- Herceptin is given intravenously and normally over an hour
- In early stage breast cancer:
 - administered every 3 weeks for a year
- Side-effects:
 - Nausea (normally lasts for 1-2 days)
 - Heart effects
 - Fatigue

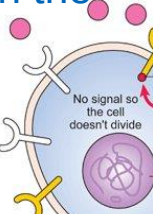


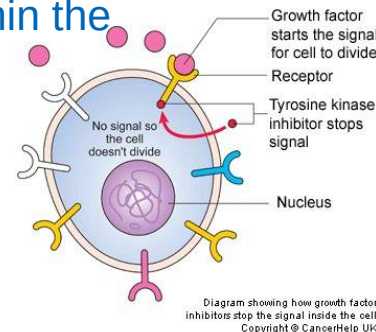
Cetuximab (Erbitux®)

- Some cancers have a large number of receptors on their surface called epidermal growth factor receptors (EGFRs). When activated they trigger cancer cells to divide and grow
- Cetuximab locks onto the EGFR's and stops them from stimulating the cancer cells to divide and grow
- Cetuximab makes the cancer cells more sensitive to chemotherapy and radiotherapy
- Cetuximab only effective on bowel cancers that have a normal KRAS gene
- Licensed for advanced (metastatic) large bowel and head and neck cancer
- Administered intravenously as a weekly hourly infusion.
- The side effects of Cetuximab are generally mild and include:
 - Nausea
 - Headaches
 - Diarrhoea
 - Fatigue
 - Skin changes – rash on face & neck - common



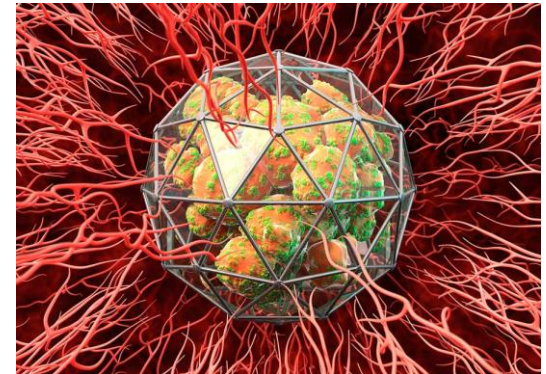
Cancer growth inhibitors

- In order to grow and divide cancer cells 'communicate' with each other using chemical signals
 - Cancer growth inhibitors interfere with this process and so affect the cancer's ability to develop.
 - Scientists are looking at different ways of doing this such as:
 - *Lowering levels of the growth factor in the body*
 - *Blocking the growth factor receptor on the cancer cell*
 - *Blocking the signals inside the cell that start up when the growth factor triggers the receptor*
 - Cancer growth blockers are named after the chemical they block
 - Example: **Lapatinib (Tyverb®)** used to treat advanced breast cancer – administered in a tablet form. Tyrosine kinase inhibitor - inhibits signals within the cancer cell that make them grow and divide
 - Side-effects: Diarrhoea (common), Skin changes, Fatigue, Nausea, Sore mouth, joint aches, poor sleep and headaches
- 
- A diagram of a cell with a purple nucleus. A yellow growth factor is binding to a receptor on the cell membrane. A red arrow points from the receptor into the cell, indicating a signal. A text box with a red arrow pointing to the receptor says "No signal so the cell doesn't divide".



Angiogenesis inhibitors

- For tumours to survive they need to produce a network of blood vessels
- Angiogenesis means the growth of new blood vessels
- Angiogenesis inhibitors interfere with the development of blood vessels, meaning the cancer is unable to receive oxygen and nutrients to survive
- Angiogenesis inhibitors may only stop or slow the growth of the cancer, they may not completely eradicate the cancer
- Two common angiogenesis inhibitors include:**
 - Lenalidomide** – used to treat myeloma
 - Thalidomide** – used to treat myeloma



Radiotherapy

- Radiotherapy is the use of high-energy rays, usually x-rays and similar rays (such as electrons) to treat disease. It works by destroying cancer cells in the area that's treated
- Radiotherapy treatment works by damaging the DNA within the cancer cells
- Two ways of administering of radiotherapy: external beam and internal radiotherapy
- About 4 out of 10 people with cancer (40%) have radiotherapy as part of their treatment
- Aim of radiotherapy is to destroy and cure the cancer or in palliative radiotherapy to relieve symptoms.



Side-effects of radiotherapy

- External radiotherapy tends to cause more general side-effects than internal therapy
- Most side-effects continue for 10-15 days after treatment is complete and then start to improve
- Main side-effects:
 - Fatigue
 - Nausea
 - Skin reactions
 - Hair loss
- Long term side-effects

